

BAAZ & UDAY JKBOSE

2026 CLASS 11TH

GRAVITATION

LECTURE-2

PHYSICS

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Topics To Be Covered

- 1 Revision (dec-1) + Variation of g with depth
- 2 Gravitational Potential Energy
- 3 Gravitation Potential
- 4 Kepler Law of Planetary Motion
- 5 Escape Velocity
- 5 Orbital Velocity

Recap From 1st Lecture

Universal Law of Gravitation

$$F = G \frac{m_1 m_2}{r^2} \quad (m_1) \text{---} r \text{---} (m_2)$$

$$G = 6.67 \times 10^{-11} \frac{\text{Nm}^2}{\text{kg}^2}$$

Vector form of Coulomb Law

$$F = \frac{G m_1 m_2}{(r_2 - r_1)^3} (\vec{r}_2 - \vec{r}_1) \quad \text{JKBOSE}$$

Acceleration due to Gravity



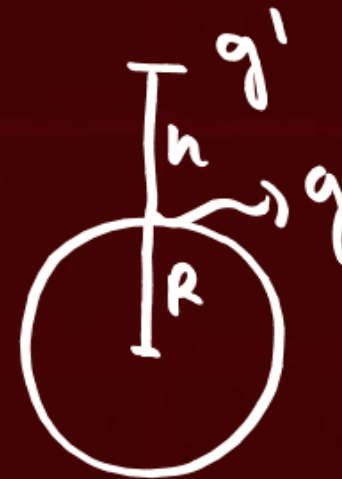
$$g = \frac{GM}{r^2}$$

$$g = G \frac{4}{3} \pi r \rho$$

$$g = -10 \text{ m/s}^2$$

$$g = -9.8 \text{ m/s}^2$$

Variation of g with height



$$g' = g \left(1 - \frac{2h}{R}\right)$$

$$g > g' \checkmark$$

① Variation of (g) with depth

- let us consider the earth with mass (m) and Radius (R)

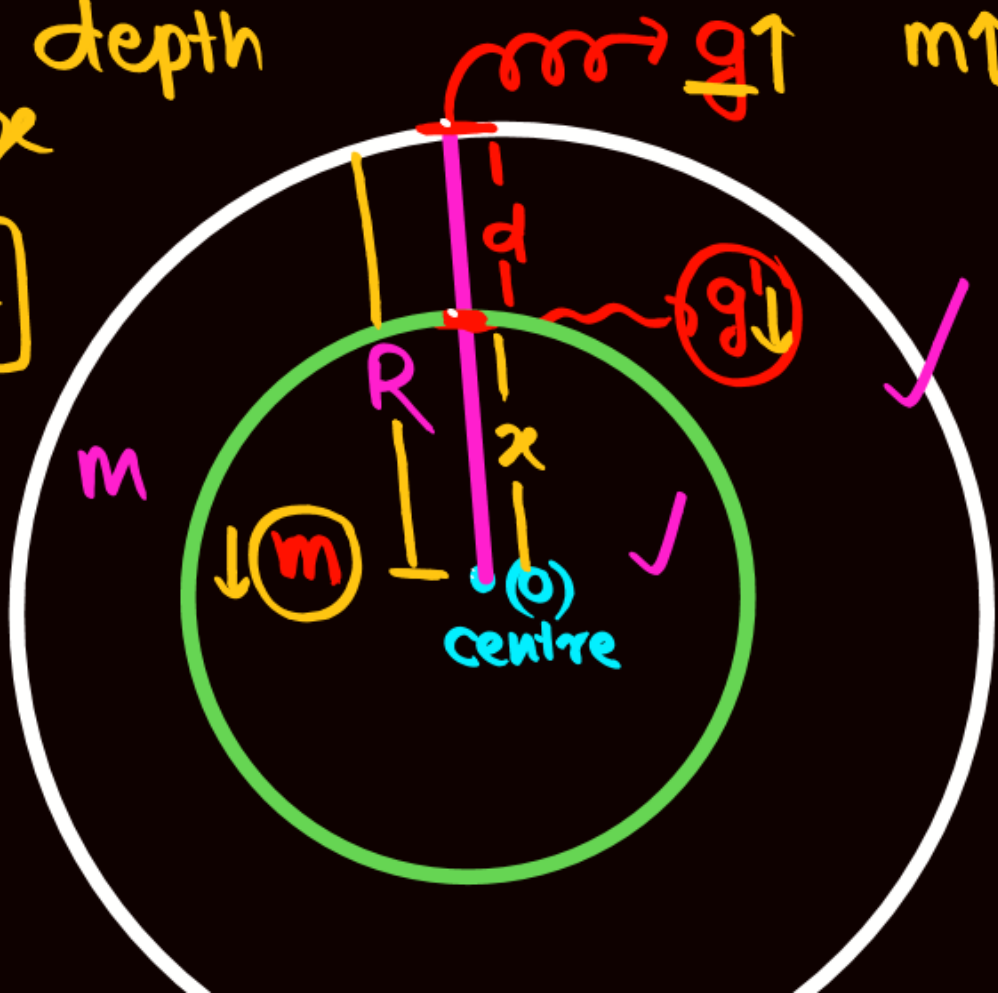
$g \Rightarrow$ acc. due to gravity at Surface

$g' \Rightarrow$ acc. due to gravity at depth

$$R = d + x$$

$$R - d = x$$

$$x = R - d$$



$$\uparrow \downarrow g = \frac{GM}{R^2} \downarrow \quad \times \quad g' < g$$

(Density wali formula)
lagane ga ✓

on surface

$$\uparrow g = \frac{4}{3} \pi G \rho R \uparrow \quad \text{--- (I)}$$

at depth

$$\downarrow g' = \frac{4}{3} \pi G \rho x \downarrow$$

$$g' = \frac{4}{3} \pi G \rho (R - d) \quad \text{--- (II)}$$

Divide --- (II) and (I)

$$\frac{g'}{g} = \frac{\cancel{\frac{4}{3}} \pi \cancel{G} \rho (R - d)}{\cancel{\frac{4}{3}} \pi \cancel{G} \rho R}$$

$$\frac{g'}{g} = \frac{(R-d)}{R}$$

$$\frac{g'}{g} = \frac{(R' - d \frac{R'}{R})}{R}$$

$$\frac{g'}{g} = \frac{R \left(1 - \frac{d}{R}\right)}{R}$$

$$\frac{g'}{g} = \left(1 - \frac{d}{R}\right)$$

$$g' = g \left(1 - \frac{d}{R}\right)$$

$$g' = g - g \frac{d}{R}$$

$$g' = 10 - 2$$

$$g' = 8$$

$$g \gg g'$$

Q numerical

JKBOSE-2017 ✓

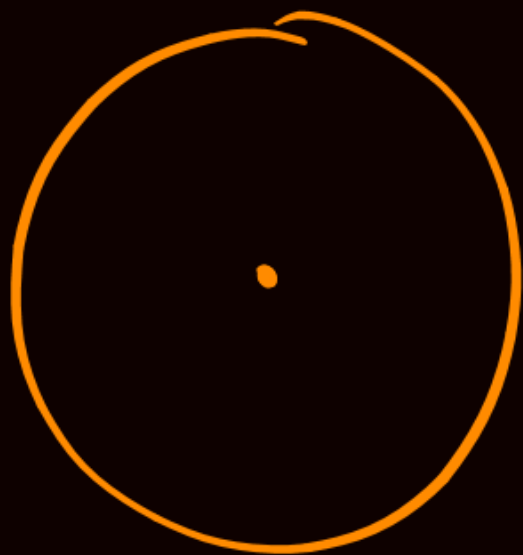
At the centre of earth value of g ?

(a) 10

(b) 9.8

(c) 4.8

(d) 0



$$g = \frac{GM}{R^2} = 0$$

$$g = \frac{G \times 0}{R^2}$$

$$g = \frac{0}{R^2} = 0 \checkmark$$

③ Shape of the Earth

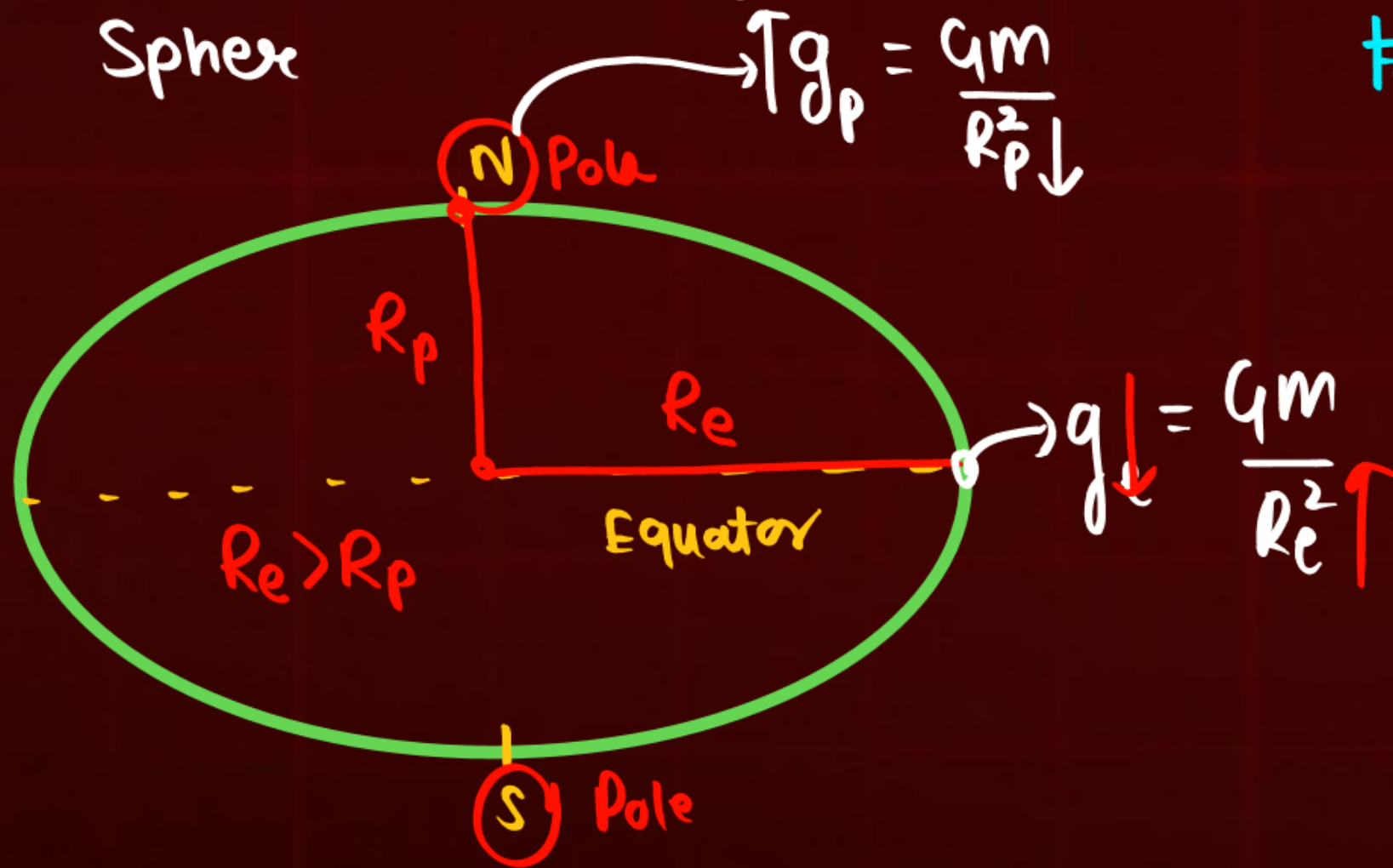
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2024

$$g = \frac{Gm}{R^2}$$

- As all we know the shape of earth is not perfectly sphere

$$\# g_{\text{pole}} \uparrow \uparrow \quad R_p \downarrow \downarrow$$

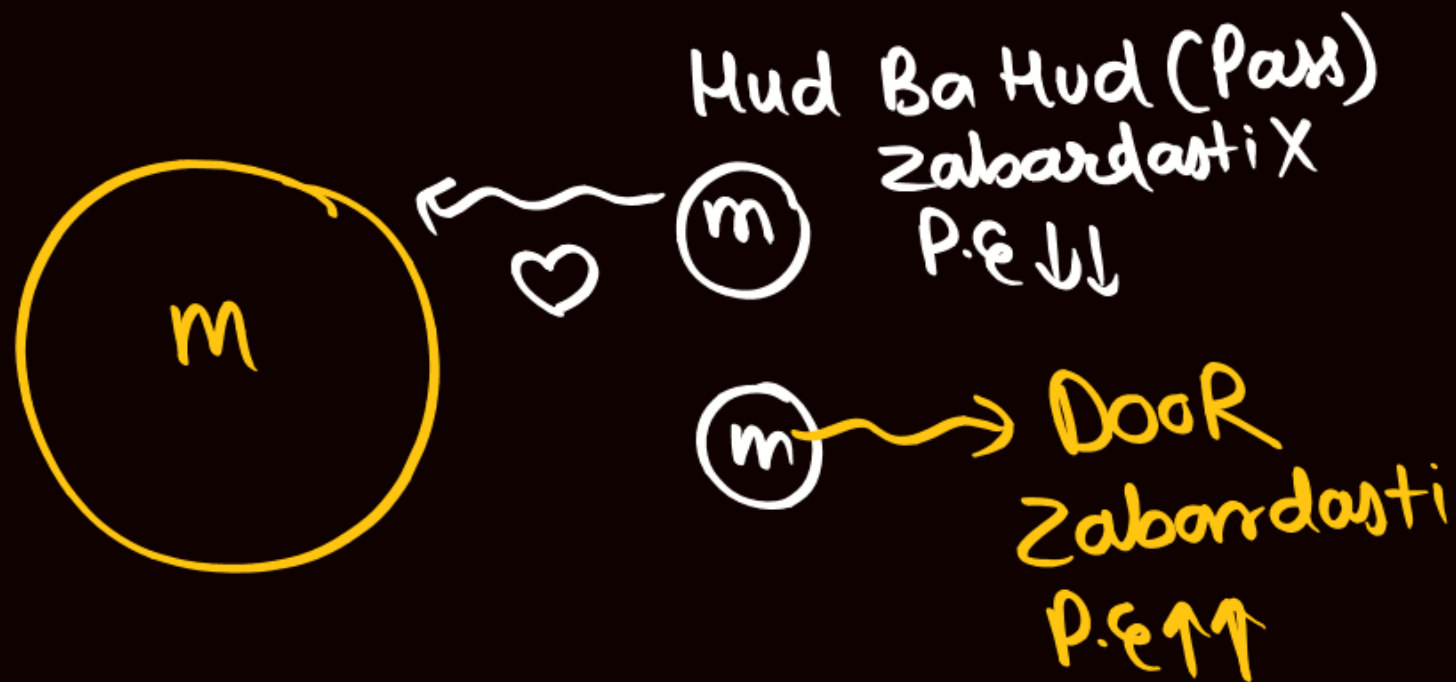
$$\# g_{\text{equator}} \downarrow \downarrow \quad R_e \uparrow \uparrow$$



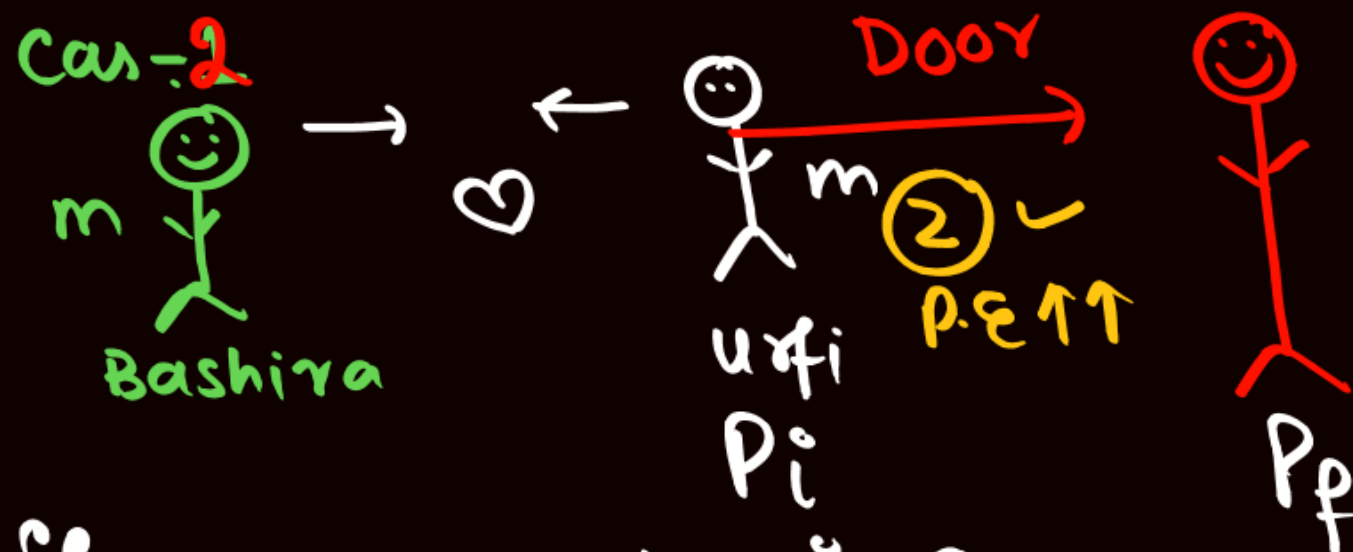
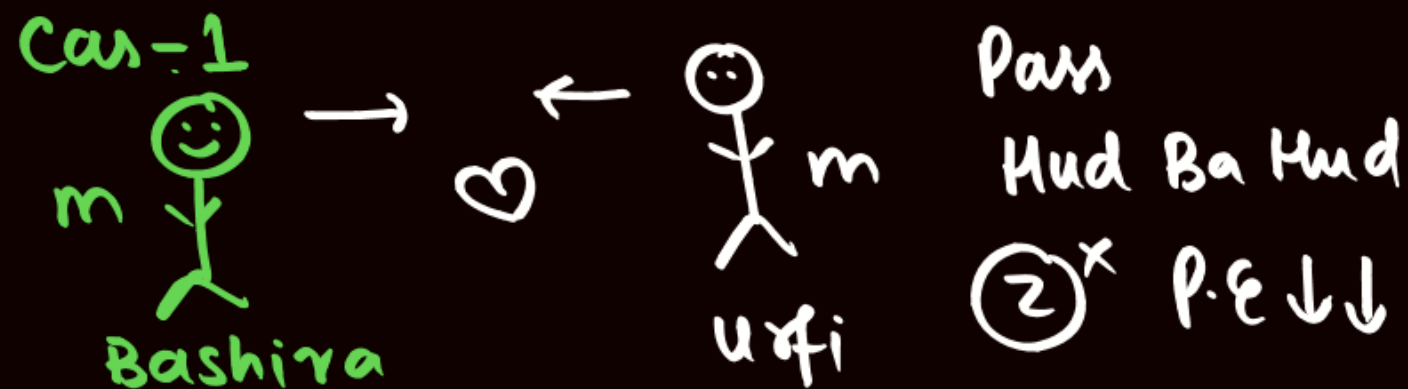
Potential Energy

⇒ Energy store due to Zabardasti

① Gravitation potential Energy



The Desi Example



“ The Energy store in a particle due to relative position of a particle

$$P.E / U \Rightarrow \underline{\text{WORK}}$$

Gravitational Potential Energy

- P.E. Store in a Mass due to the Relative position of mass

JKBOSE → 2024
MCO ✓

Scalar

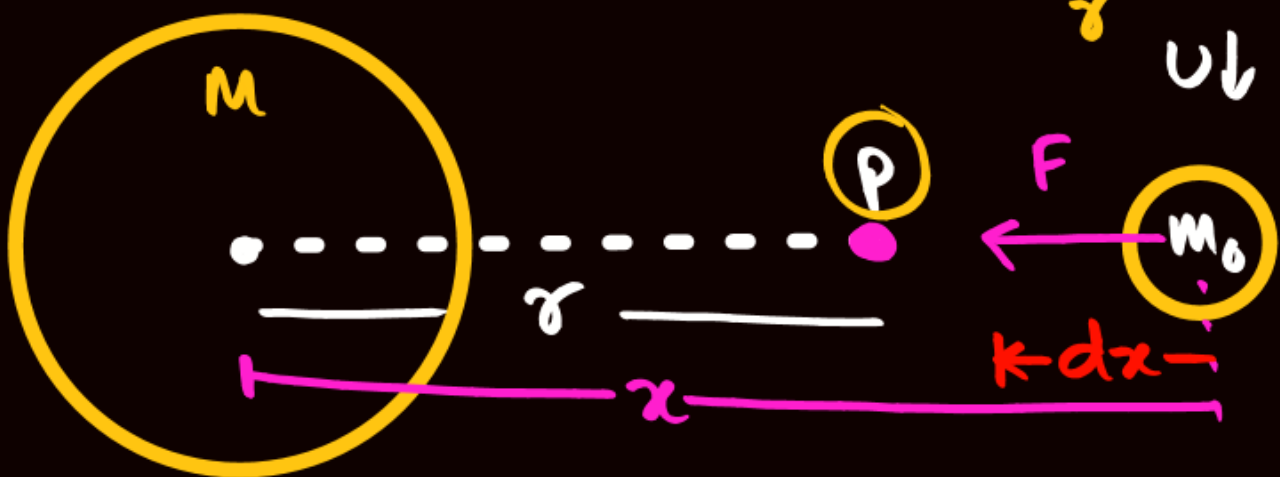
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OR

- Gravitational potential Energy is work done in bring unit mass from ∞ to that particular Point (P)

$$U = -\frac{GMm_0}{r} \quad \text{(JKBOSE)} \\ 2024 \\ 2023 \\ 2022$$

Derivation



$$F = \frac{GMm_0}{x^2} \text{ (variable force)}$$

$$W = \vec{F} \cdot \vec{x}$$

$$dW = F dx$$

$$dW = \frac{GMm_0}{x^2} dx$$

∞
 m_0
ut fi

$$m_0 = 1 \text{ kg}$$

$$dW = \frac{GMm_0}{x^2} dx$$

on integrating b/s

$$\int dW = \int_{\infty}^P \frac{GMm_0}{x^2} dx$$

$$W = GMm_0 \int_{\infty}^r \frac{1}{x^2} dx$$

$$W = GMm_0 \int_{\infty}^r x^{-2} dx$$

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$$W = G M m_0 \gamma \int_{\infty}^r x^{-2} dx$$

$$\# \int x^n dx = \frac{x^{n+1}}{n+1}$$

$$\frac{1}{\infty} = 0$$

$$W = G M m_0 \left[\frac{x^{-2+1}}{-2+1} \right]_{\infty}^r \Rightarrow \text{limit wali integration}$$

$$W = G M m_0 \left[\frac{x^{-1}}{-1} \right]_{\infty}^r$$

$$W = \frac{G M m_0}{-1} \left[x^{-1} \right]_{\infty}^r$$

$$W = -G M m_0 \left[\frac{1}{x} \right]_{\infty}^r$$

$$W = -G M m_0 \left[\frac{1}{x} \right]_{\infty}^r$$

$$W = -G M m_0 \left[\frac{1}{r} - \cancel{\frac{1}{\infty}} \right]_0$$

$$W = -\frac{G M m_0}{r}$$

$$W = - \frac{G m m_0}{r}$$

$$U = - \frac{G m m_0}{r}$$

↓

U ↓

Gravitational potential (V)

the Gravitational P.E
per unit mass

$$V = \frac{U}{m_0}$$

$$V = \frac{U}{m_0} = \frac{J}{kg}$$

HLW

$$V = \frac{-\frac{GMm_0}{r}}{m_0} = -\frac{GM}{r}$$

Kya ye easy hai?

H/W



At what depth from the surface of the earth the value of acceleration due to gravity would become half of its value on the surface of the earth? Radius of the earth is 6400 km.

Can You Solve it?

M/w

M

U=?

If the mass of the earth is 6×10^{24} kg. then find the potential energy of an object of mass 33.5 kg at a distance of 3.35×10^{10} m from the centre of the earth. $G = 6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$

$$G = 6.67 \times 10^{-11} \frac{\text{N m}^2}{\text{kg}^2}$$



Samjho, Feel Karo

Mastery

Apne Ap Hogi

Thank
You

